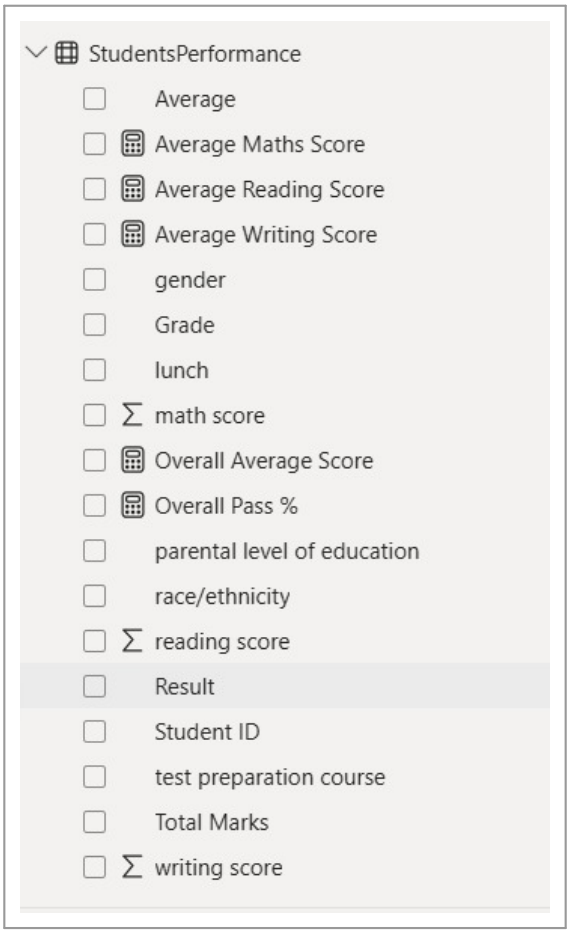
Week 7 Project Report

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1. StudentsPerformance Table – Fields Pane



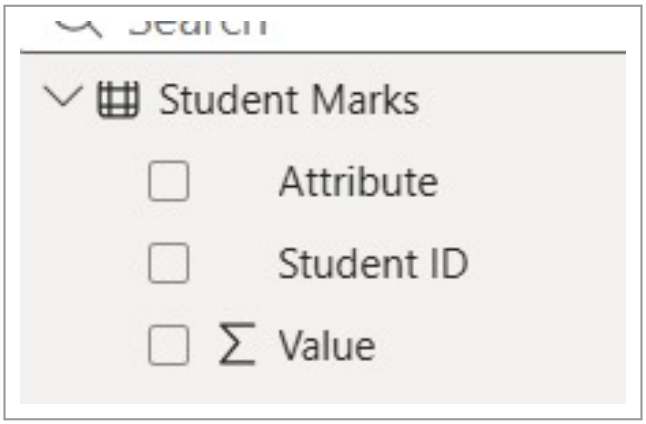
This screenshot shows the Fields pane of the StudentsPerformance table, which is the main dataset used in the project. It holds each student's demographic details and exam scores, along with calculated columns and DAX measures created for analysis.

Original data columns: gender, race/ethnicity, parental level of education, lunch, test preparation course, math score, reading score and writing score. The score columns carry the Σ (sum) symbol, showing they are numeric fields that can be aggregated.

Calculated columns: Average, Grade, Result and Total Marks. These derive each student's total and average marks, assign a grade, and mark the student as Pass or Fail.

DAX measures (calculator icon): Average Maths Score, Average Reading Score, Average Writing Score, Overall Average Score and Overall Pass %. These measures supply the values shown in the KPI cards of the dashboard.

2. Student Marks Table



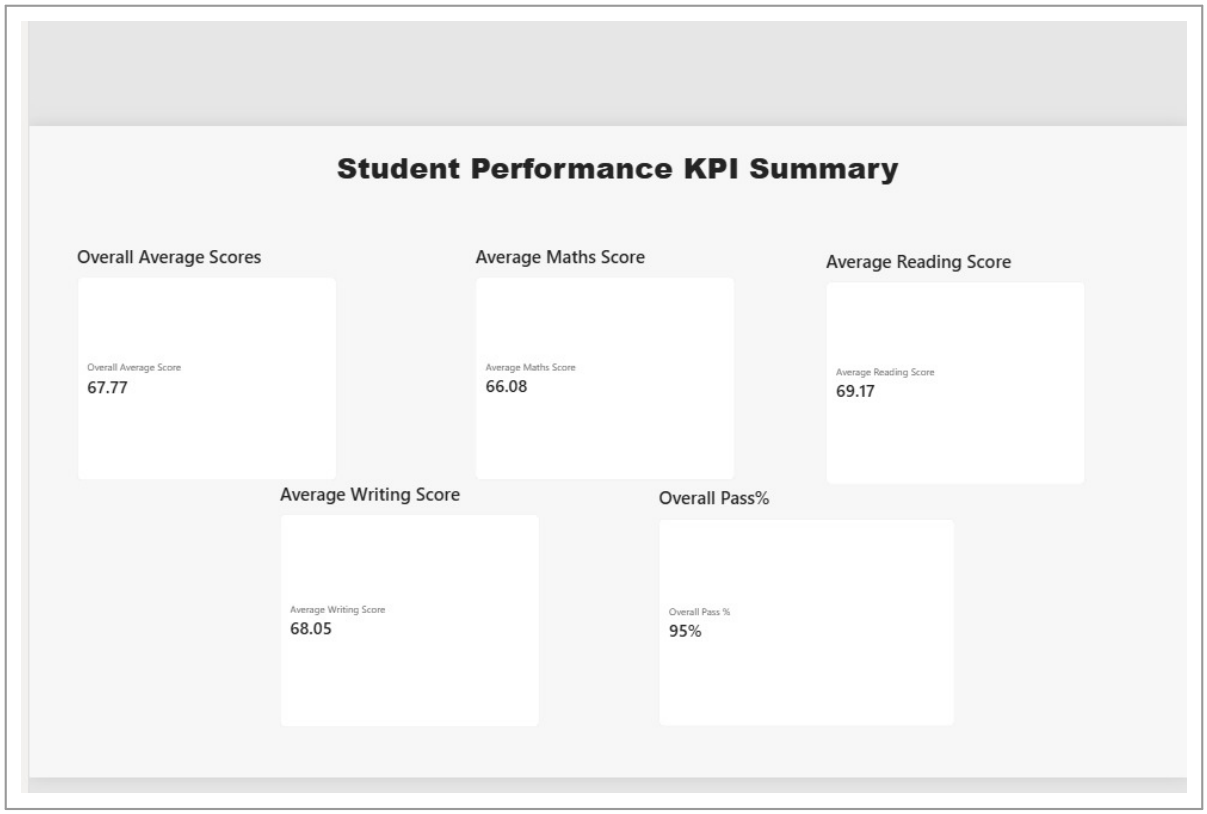
This screenshot shows a second table named Student Marks, containing three fields: Attribute, Student ID and Value.

This is the unpivoted (long) format of the data, typically produced in Power Query using the "Unpivot Columns" option. Instead of separate Math, Reading and Writing columns, each subject becomes a row:

- Attribute – the subject name (e.g., math score, reading score, writing score).
- Student ID – the unique identifier linking each record to a student.
- Value – the marks scored in that subject (numeric, summed with Σ).

This structure makes it easier to build visuals that compare subjects side by side, such as a single chart with Attribute on the axis and Value as the measure, and it works well with slicers and filters.

3. Student Performance KPI Summary Dashboard



This is the final report page, titled Student Performance KPI Summary. It uses five KPI cards to present the key results at a glance:

Overall Average Score (67.77): the mean of all subject scores across students, giving a general picture of performance.

Average Maths Score (66.08): the average marks in Mathematics, the lowest of the three subjects.

Average Reading Score (69.17): the average marks in Reading, the highest of the three subjects.

Average Writing Score (68.05): the average marks in Writing, which sits between Maths and Reading.

Overall Pass % (95%): the share of students who passed, showing very strong overall results.

Reading has the highest average (69.17), while Maths is the lowest (66.08), pointing to Maths as the subject needing the most support.

4. Conclusion

In this Week 7 project, a student performance dataset was prepared and analysed in Power BI. The StudentsPerformance table was enriched with calculated columns (Average, Grade, Result, Total Marks) and DAX measures, while the Student Marks table, created by unpivoting the subject columns, provided a flexible structure for subject-wise comparison.

The KPI Summary dashboard brings the results together in a clear, readable form. The overall average score of 67.77 and a pass percentage of 95% show that students performed well overall. Among the subjects, Reading (69.17) is strongest, followed by Writing (68.05) and Maths (66.08).

The project demonstrates key Power BI skills: data modelling, creating calculated columns and DAX measures, transforming data with Power Query, and designing a KPI report that supports data-driven decisions in education. Future work could add visuals by gender, parental education and test preparation course to explore what influences performance.